

Problem Statement: AI-Powered Menopause Wellness Agent

The Challenge

Menopause is a natural transition that affects millions of women worldwide, yet it remains underrecognized and poorly supported. Women often experience physical, emotional, and cognitive symptoms during pre-menopause, perimenopause, menopause, and post-menopause, making it difficult to understand their current stage and manage their health effectively. Reliable, evidence-based information is scattered across multiple sources, and most existing solutions provide generic advice rather than personalized guidance. Without continuous, individualized support, women may struggle with symptom management, nutrition, mental well-being, fitness, sleep, and workplace productivity.

Objectives

Educate & Assess: Help women understand menopause and identify their current stage (pre-, peri-, menopause, or post-menopause).

Personalized Wellness: Provide tailored recommendations for nutrition, exercise, sleep, mental well-being, and self-care.

Continuous Support: Enable symptom tracking, proactive health insights, and preventive care guidance.

Empower Women: Promote informed decision-making, healthier lifestyles, and improved workplace well-being through trusted, evidence-based AI support.

Proposed Solution: AI-Powered Menopause Wellness Agent

The Objective

Build an AI-powered Menopause Wellness Agent that delivers personalized, evidence-based guidance to support women throughout their menopause journey. The solution should include:

Menopause Stage Assessment – Help users identify whether they are in pre-menopause, perimenopause, menopause, or post-menopause based on symptoms, age, menstrual history, and lifestyle.

Personalized Wellness Plans – Recommend daily nutrition, exercise, sleep, stress management, hydration, and self-care routines tailored to individual health conditions, preferences, and goals.

Evidence-Based Knowledge RAG Retrieval – Retrieve and summarize trusted medical information from reliable sources such as WHO, NIH, NAMS, and peer-reviewed clinical guidelines using Retrieval-Augmented Generation (RAG).

Symptom Tracking & Personalized Feedback – Allow users to log symptoms, mood, sleep quality, and activity through text or voice, providing real-time insights and personalized recommendations.

Preventive Health Support – Offer guidance for maintaining bone health, cardiovascular health, hormonal balance, mental well-being, and preventive screenings while encouraging timely consultation with healthcare professionals when appropriate.

Multi-Agent System

Menopause Knowledge Agent (retrieves evidence-based menopause information using RAG)

Stage Assessment Agent (determines the user's likely menopause stage)

Wellness Recommendation Agent (creates personalized diet, fitness, sleep, and lifestyle plans)

Symptom Monitoring & Feedback Agent (tracks symptoms, identifies trends, and provides adaptive recommendations)

Visualization Dashboard – Display symptom trends, wellness scores, sleep quality, nutrition adherence, activity levels, and progress toward personal health goals through intuitive charts and reports.

Technology Used:

IBM watsonx Orchestrate (IBM Bob): Multi-agent workflow design and orchestration.

IBM Granite 4.0 Small Instruct: Natural language understanding, reasoning, and personalized recommendations.

IBM Cloud: Secure, scalable infrastructure for deployment and data management.

IBM watsonx.ai: Model deployment, embeddings, vector search, and AI governance.

RAG (Retrieval-Augmented Generation): Retrieves trusted menopause knowledge before generating personalized responses.

Vector Database (FAISS/Chroma): Stores medical knowledge embeddings for fast, context-aware retrieval.

Trusted Data Sources: WHO, NIH, The Menopause Society (NAMS), ACOG, and peer-reviewed clinical guidelines.